

2017

Trial
Higher School
Certificate
Examination



MATHEMATICS EXTENSION 1

21 July 2017

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen.
- NESA approved calculators may be used.
- Commence each new question in a new booklet. Write on both sides of the paper.
- A reference sheet is provided.
- In Question 11-14 show relevant mathematical reasoning and/or calculations
- At the conclusion of the examination, bundle the booklets used in the correct order **including your reference sheet** within this paper and hand to examination supervisor.

Total Marks: Section 1 – 10 marks (pages 2 - 5)

70

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section 2 – 60 marks (pages 6 - 10)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

NESA NUMBER:

BOOKLETS USED:

Marker's use only.

QUESTION	1-10	11	12	13	14	Total
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$

This task constitutes 40 % of the HSC Course Assessment

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 11)

1. $\int \sin 2x \, dx =$

I $\sin^2 x + C$ where C is a constant

II $-\frac{1}{2} \cos 2x + D$ where D is a constant

III $-\cos^2 x + E$ where E is a constant

Which of above solutions is correct?

(A) II only

(B) I and II only

(C) II and III only

(D) I, II and III

2. In how many ways can a committee of 2 men and 3 women be selected from a group of 6 men and 8 women?

(A) ${}^6P_2 \times {}^8P_3$

(B) ${}^6C_2 \times {}^8C_2$

(C) ${}^6P_3 \times {}^8C_2$

(D) ${}^6C_2 \times {}^8C_3$

3. What are the solutions to the equation $e^{6x} - 7e^{3x} + 6 = 0$?
- (A) $x = 1$ and $x = 6$
- (B) $x = 0$ and $x = \frac{\log_e 6}{2}$
- (C) $x = 0$ and $x = \frac{\log_e 6}{3}$
- (D) 6
4. If P is the point $(-1, 2)$ and K is the point $(3, 5)$, the coordinates of the point that divides the interval PK externally in the ratio $2 : 3$ are:
- (A) $(11, 11)$
- (B) $\left(\frac{3}{5}, \frac{16}{5}\right)$
- (C) $(-3, -4)$
- (D) $(-9, -4)$

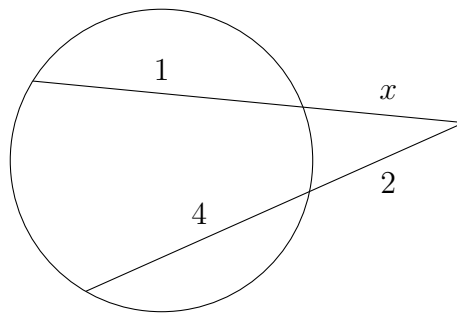
5.

$$\lim_{x \rightarrow 0} \frac{\sin \frac{3x}{2}}{6x} =$$

- (A) $\frac{1}{4}$
- (B) 4
- (C) $\frac{1}{2}$
- (D) 2

6. For the functions $f(x) = \frac{1}{x}$ and $g(x) = 2 \sin x$ the range of $g(f(x))$ is
- (A) $-1 \leq y \leq 1$
- (B) $\frac{1}{2} \leq y \leq \frac{1}{2}$
- (C) $-1 \leq y \leq 1$
- (D) $-2 \leq y \leq 2$
7. What is the exact value of $\tan \left(\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) \right)$?
- (A) $\frac{1}{\sqrt{3}}$
- (B) $-\frac{1}{\sqrt{3}}$
- (C) $\sqrt{3}$
- (D) $-\sqrt{3}$
8. $\cos^{-1} [\cos(\cos^{-1}(-x))]$
- (A) $\pi - x$
- (B) $\pi - \cos^{-1} x$
- (C) $\cos^{-1} x$
- (D) $-\cos^{-1} x$
9. A parabola has the parametric equation $x = 6t$, $y = 3t^2$. The Cartesian equation of its derivative is:
- (A) $y' = 6t$
- (B) $y' = \frac{x^2}{12}$
- (C) $y' = \frac{x}{6}$
- (D) $y' = \frac{2x}{9}$

10. Find the value of x .



NOT TO SCALE

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$

Section II

70 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Use a SEPARATE writing booklet	Marks
(a) Solve the inequality $\frac{5}{x-2} \leq 4$		3
(b) Use the substitution $u = x+2$ to evaluate $\int_1^3 \frac{x+1}{(x+2)^2} dx$ giving the answer in simplest exact form.		3
(c) The polynomial $P(x) = 2x^3 + bx^2 - 8x + 5$ has a remainder of 2 when divided by $(x+3)$, calculate the remainder when divided by $(x+2)$.		3
(d) i. Show that that $y = e^{-x} - 3x$ has only one real root.		1
ii. The root lies in the interval $0.2 < x < 0.3$ Taking $x_1 = 0.25$ as the first approximation, use one application of Newton's method to determine a better approximation. Give your answer correct to 3 significant figures.		2
(e) Using the word COWABUNGA		
i. How many arrangements of the 9 letters are possible?		1
ii. How many arrangements of the 9 letters are possible if the 3 letters COW have to be placed together? Note the letters COW can be placed in any order but they must be together.		2

Question 12 (15 Marks)

Use a SEPARATE writing booklet

Marks

- (a) The polynomial $P(x) = x^3 - 3x^2 + kx + 48$ has roots α, β and γ **3**

Two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of k .

- (b) A particle whose displacement is x moves in simple harmonic motion. Given that the displacement equation is of the form $x = a \cos(nt + \alpha)$ and $\ddot{x} = -4x$.

Initially the particle has displacement of $x = \sqrt{3}$ and velocity of $\dot{x} = 6$.

- i. Find the value of n . **1**

- ii. Find the displacement equation. **3**

- (c) Determine the domain and range of $y = \frac{1}{2} \sin^{-1}(1 - 3x)$ and hence sketch the graph. **3**

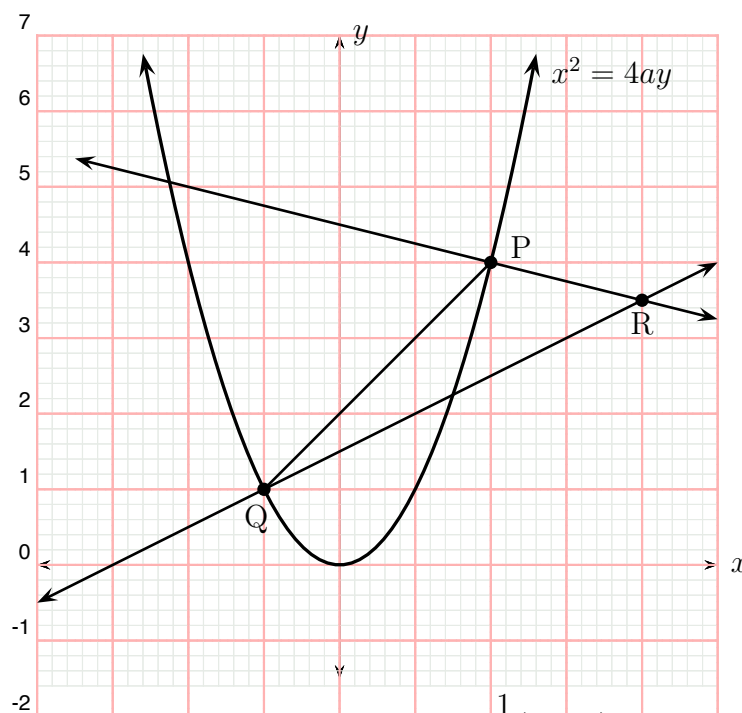
Question 12 continues on page 8

Question 12 (continued)

- (d) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ as shown below.

The equation of the normal to the parabola at P is $x + py = 2ap + ap^3$.

The equation of the normal to the parabola at Q is $x + qy = 2aq + aq^3$.



- i. The equation of the chord PQ is $y = \frac{1}{2}(p+q)x - apq$. (DO NOT show this). If the chord passes through $(0, 5a)$ show that $pq = -5$. 1
- ii. Show that the point of intersection, R , of the normals at P and Q has the coordinates $[-apq(p+q), a(p^2 + pq + q^2 + 2)]$. 2
- iii. Find the equation of the locus of R if the chord PQ passes through $(0, 5a)$. 2

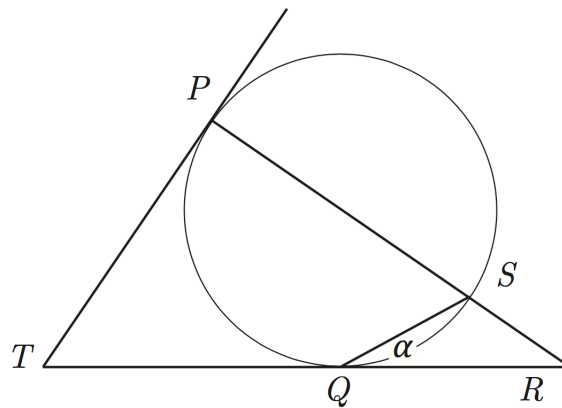
Question 13 (15 Marks)

Use a SEPARATE writing booklet

Marks

- (a) i. Show that $y = \frac{x^2 + x + 1}{x + 1} = x + \frac{1}{x + 1}$ **1**
- ii. Sketch the graph of the function, clearly showing all key features **3**

- (b) PS is a diameter of the circle PSQ . The tangents at P and Q meet at T . The tangent TQ and the diameter PS are produced to meet at R . Let $\angle SQR = \alpha$.



- i. Prove that $\angle TPQ = \frac{\pi}{2} - \alpha$. **2**
- ii. Prove that $\angle PTQ = 2\alpha$. **2**
- (c) The acceleration of a body A is given by $a = 18x(x^2 + 1)$ where x cm is the displacement after time t seconds. Initially, A starts from the origin with velocity 3 cm s^{-1} .
- i. Show that $v = 3(x^2 + 1)$. **2**
- ii. Find x in terms of t . **2**
- (d) By writing $\sqrt{3} \sin 3x - \cos 3x$ in the form $R \sin(3x - \alpha)$, where α is acute and $R > 0$. or otherwise, find the general solutions to the equation:

$$\sqrt{3} \sin 3x - \cos 3x = 0$$

Question 14 (15 Marks)

Use a SEPARATE writing booklet

Marks

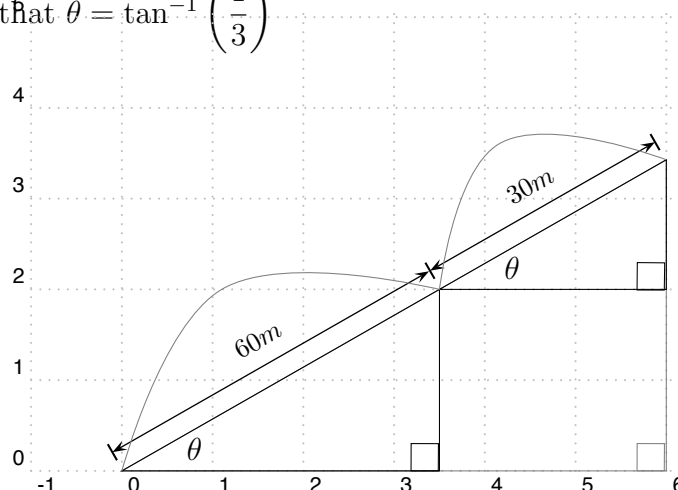
- (a) A boy throws a ball with velocity $v \text{ ms}^{-1}$ at an angle of 45° to the horizontal.

- i. Derive expressions for the horizontal and vertical displacement of the ball from the point of projection. **2**
- ii. Show that the Cartesian equation of the path of the ball is **2**

$$y = x - \frac{gx^2}{v^2}$$

where g is the acceleration due to gravity.

- iii. A boy is standing on a hill inclined at θ to the horizontal as shown in the diagram below. He throws the ball at the same angle of elevation of 45° and at the same speed of $v \text{ ms}^{-1}$. If he can throw the ball 60 m down the hill but only 30 m up the hill, use the results in part ii. to show that $\theta = \tan^{-1}\left(\frac{1}{3}\right)$ **3**



- (b) The surface area of a spherical bubble is increasing at a constant rate of 1.9 mm^2 per second. Find the rate of increase of its volume when the radius is 0.6 mm . The volume of a sphere is given by $V = \frac{4}{3}\pi R^3$, and the surface area is given by $A = 4\pi R^2$. **3**

- (c) i. Using the expansions of $\cos(A+B)$ and $\cos(A-B)$, show that: **2**

$$\cos x - \cos(x+2\theta) = 2 \sin(x+\theta) \sin \theta$$

- ii. Hence prove by mathematical induction that **3**

$$\sum_{r=1}^n \sin(2r-1)\theta = \frac{1 - \cos(2n\theta)}{2 \sin \theta}$$

End of Examination ☺



MATHEMATICS EXTENSION 1

NESA NUMBER:

Section 1 – Multiple Choice Answer Sheet

Use this multiple choice answer sheet for Questions 1 - 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

- Start Here** →
1. A ☐ B ☐ C ☐ D ☐
 2. A ☐ B ☐ C ☐ D ☐
 3. A ☐ B ☐ C ☐ D ☐
 4. A ☐ B ☐ C ☐ D ☐
 5. A ☐ B ☐ C ☐ D ☐
 6. A ☐ B ☐ C ☐ D ☐
 7. A ☐ B ☐ C ☐ D ☐
 8. A ☐ B ☐ C ☐ D ☐
 9. A ☐ B ☐ C ☐ D ☐
 10. A ☐ B ☐ C ☐ D ☐

Multiple Choice Answer Sheet

Student Number Ext 1 - Answers - 2017 Trial.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 ↑ *correct*

1. A ☐ B ☐ C ☐ D ☒
2. A ☐ B ☐ C ☐ D ☒
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☒ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☒
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☐ B ☐ C ☒ D ☐
10. A ☐ B ☐ C ☒ D ☐

Ext 1 - Multiple Choice Answers - 2017 Trial.

1 $y = (\sin x)^2$
 $y' = 2 \sin x \cos x = \sin 2x$

$$y = -\frac{1}{2} \cos 2x$$
$$y' = -\frac{1}{2} \times 2x - \sin 2x = -\sin 2x$$

$$y = -\cos^2 x = -(\cos x)^2$$
$$y' = 2x - x - \sin x \times \cos x$$
$$= 2 \sin x \cos x = \sin 2x$$

$\therefore$ (D)

2 need 2 men from 6, 3 women from 8

$$\therefore {}^6C_2 \times {}^8C_3 \quad \text{it is C not P as a committee}$$

$\therefore$ (D) (order is unimportant)

3 $\rightarrow$ you can sub. in values + test

OR $e^{6x} - 7e^{3x} + 6 = 0$

let $u = e^{3x} \quad \therefore u^2 - 7u + 6 = 0$
 $(u-6)(u-1) = 0$

$\therefore u = 6, 1$

when $u = 1$

$$e^{3x} = 1$$

$$\ln(e^{3x}) = \ln(1)$$

$$\therefore 3x = 0$$

$$\therefore x = 0$$

when $u = 6$

$$e^{3x} = 6$$

$$\ln(e^{3x}) = \ln(6)$$

$$\therefore 3x = \ln(6)$$

$$\therefore x = \frac{\ln(6)}{3}$$

$\therefore$ (C)

$$\underline{4} \quad \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$\begin{array}{ccc} (-1, 2) & (3, 5) & (-2, 3) \\ (x_1, y_1) & (x_2, y_2) & (m, n) \end{array}$$

$$= \left(\frac{-2 \times 3 + 3 \times -1}{1}, \frac{-2 \times 5 + 3 \times 2}{1} \right)$$

$$= (-9, 4) \therefore (D)$$

$$\underline{5} \quad \lim_{x \rightarrow 0} \frac{\sin \frac{3x}{2}}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{4} \left(\frac{\sin \frac{3x}{2}}{\frac{6x}{4}} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{4} \left(\frac{\sin \frac{3x}{2}}{\frac{3x}{2}} \right) \quad \left(\text{as } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$= \frac{1}{4} \therefore (A)$$

$$\underline{6} \quad g[f(x)] = 2 \sin\left(\frac{1}{x}\right)$$

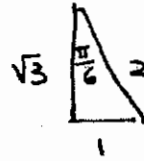
as $\frac{1}{x}$ has domain all real x
except $x \neq 0$

$\therefore$ this is just like $2 \sin x$ in terms of range

$$\sin x \quad -1 \leq y \leq 1$$

$$2 \sin x \quad \underline{-2 \leq y \leq 2} \therefore (D)$$

7 $\tan \left[\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) \right]$



$$\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$$

$$\therefore \tan \left(\frac{\pi}{6} \right) = \frac{1}{\sqrt{3}} \quad \therefore (A)$$

8 $\cos^{-1} [\cos(\cos^{-1}(x))]$

$$= \cos^{-1} [\cos(\pi - \cos^{-1}(x))]$$

$$= \cos^{-1} [\cos \pi - x]$$

$$= \pi - \cos^{-1}(x) \quad \therefore (B)$$

9 $x = 6t, y = 3t^2$

$$t = \frac{x}{6}$$

$$y = 3 \times \left(\frac{x}{6} \right)^2 = \frac{3x^2}{36}$$

$$y = \frac{x^2}{12}$$

$$y' = \frac{2x}{12} = \frac{x}{6} \quad \therefore (C)$$

10 $x(x+1) = 2(2+4)$

$$x^2 + x = 12$$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

$$\therefore x = -4, 3 \quad \text{as } x \neq 0$$

$$\therefore x = 3 \quad \therefore (C)$$

Question 11 Solutions

(a) Solve the inequality $\frac{5}{x-2} \leq 4$

3

3 marks answer and excluding $x=2$

2 marks correct answer not excluding $x=2$

1 mark correct expression with $(x-2)$ factor

$$5(x-2) \leq 4(x-2)^2$$

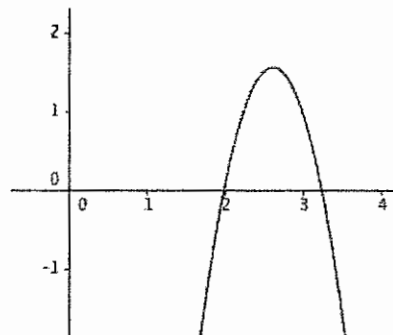
$$5(x-2) - 4(x-2)^2 \leq 0$$

$$(x-2)[5-4x+8] \leq 0 \quad (1)$$

$$(x-2)(13-4x) \leq 0$$

$$x \leq 2 \quad x \geq \frac{13}{4} \quad (2)$$

$$x < 2 \text{ as } x \neq 2 \quad x \geq \frac{13}{4} \quad (3)$$



Question was done well. There was no penalty if the solution was not graphed or tested

- (b) Use the substitution $u = x+2$ to evaluate $\int_1^3 \frac{x+1}{(x+2)^2} dx$ giving the answer in simplest exact form. 3

$$u = x + 2$$

terminals when $x = 1$ $u = 3$

$$\frac{du}{dx} = 1$$

$$x = 3 \quad u = 5$$

$$du = dx$$

$$x+1 = u-1$$

$$\int_3^5 \frac{(u-1)}{u^2} du$$

$$= \int_3^5 \frac{1}{u} - u^{-2} du$$

$$= \left[\ln u - \frac{u^{-1}}{-1} \right]_3^5$$

$$= \left[\ln u + \frac{1}{u} \right]_3^5$$

$$= \left(\ln 5 + \frac{1}{5} \right) - \left(\ln 3 + \frac{1}{3} \right)$$

$$= \ln\left(\frac{5}{3}\right) - \frac{2}{15}$$

Common error was forgetting that $\int \frac{1}{u} du = \ln u$, and not changing the terminals

Many students lost a mark for not simplifying the last line

- (c) The polynomial $P(x) = 2x^3 + bx^2 - 8x + 5$ has a remainder of 2 when divided by $(x+3)$, calculate the remainder when divided by $(x+2)$. 3

3 marks correctly value of remainder with correct working
 2 marks correct value of b
 1 mark correct substitution(-3)

$$P(-3) = 2(-27) + 9b + 24 + 5 = 2 \quad (1)$$

$$-25 + 9b = 2$$

$$9b = 27$$

$$b = 3 \quad (2)$$

$$P(x) = 2x^3 + 3x^2 - 8x + 5$$

$$P(-2) = 2(-8) + 3(4) + 16 + 5$$

$$= 17 \quad (3)$$

Question was done well

- (d) i. Show that that $y = e^{-x} - 3x$ has only one real root. 1
- ii. The root lies in the interval $0.2 < x < 0.3$ Taking $x_1 = 0.25$ as the first approximation, use one application of Newton's method to determine a better approximation. Give your answer correct to 3 significant figures. 2

(i)

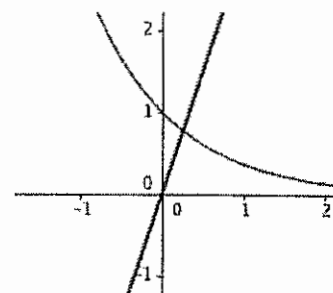
1 mark correct working showing graphs or equivalent working

$$e^{-x} - 3x = 0$$

$$e^{-x} - 3x = 0$$

$$e^{-x} = 3x$$

sketching graphs they intersect at only 1 point $\therefore$ 1 solution only



This question was done poorly. Students who graphed the function were successful.

Many students showed that $f'(x) < 0$ for all x and then stated that there were no turning points. This was not sufficient to show there is only 1 root, as there could be no roots. You must show that $f(x)$ changes sign as well.

Some students found the discriminant and showed that $\Delta < 0$. This is incorrect as the discriminant is only valid for quadratics.

(ii)

2 marks correct value of x_2 with correct working
1 mark correct derivative and substitution into the correct formula, or 1 error

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x) = e^{-x} - 3x$$

$$f'(x) = -e^{-x} - 3 \quad (1)$$

$$x_2 = 0.25 - \frac{[e^{-0.25} - 3(0.25)]}{[-e^{-0.25} - 3]}$$

$$= 0.2576....$$

$$= 0.258 \quad (2)$$

Done well. There was no penalty for incorrect rounding this time, but students need to take care.

(e) Using the word COWABUNGA

- i. How many arrangements of the 9 letters are possible? 1
- ii. How many arrangements of the 9 letters are possible if the 3 letters COW have to be placed together? Note the letters COW can be placed in any order but they must be together. 2

(i)

1 mark correct answer

$$\frac{9!}{2!} \text{ or } \frac{2262880}{2} = 181440$$

(ii)

2 marks correct answer no working required

1 mark correct answer with 1 error

$$\frac{7! \times 3!}{2!} = 15120$$

COW -----

-COW -----

--COW -----

---COW ----

----COW --

-----COW -

-----COW

$$\frac{6! \times 3! \times 7}{2} = \frac{30240}{2} = 15120$$

Done well, common error was not allowing for Double letter A

4.12 Ext 1 Marking Scheme

Q12

a

$$x^3 - 3x^2 + kx + 48$$

roots α, β, γ

$$\alpha + \beta + \gamma = \frac{-b}{a} = 3 \quad (1)$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = k \quad (2)$$

$$\alpha\beta\gamma = \frac{-d}{a} = -48 \quad (3)$$

(1 mark)

Let $\beta = -\alpha$ $\therefore$ (1) $\rightarrow \alpha + -\alpha + \gamma = 3$

$$\gamma = 3 \quad (1 \text{ mark})$$

$$(3) \quad \alpha, -\alpha, \gamma = -48$$

$$-\alpha^2 \times 3 = -48$$

$$3\alpha^2 = 48$$

$$\alpha^2 = 16$$

$$\alpha = 4, \quad -\alpha = -4$$

$\therefore$ (2) Becomes

$$\alpha \cdot -\alpha + \cancel{\alpha\gamma} - \cancel{\alpha\gamma} = k$$

$$4 \times -4 = k$$

$$\therefore k = -16$$

(1 mark)

(lose 1 mark if α is incorrect)

126

$$x = a \cos(nt + \epsilon)$$

$$\ddot{x} = -4x \quad t=0, x=\sqrt{3}, \dot{x}=6$$

i

$$\ddot{x} = -n^2 x$$

$$-4x = -n^2 x$$

$$\therefore n^2 = 4$$

$$\therefore \underline{n=2}$$

iiwhen $t=0$

$$x = a \cos \epsilon$$

$$\therefore \sqrt{3} = a \cos \epsilon \quad (1)$$

$$\text{as } x = a \cos(nt + \epsilon)$$

$$\dot{x} = -na \sin(nt + \epsilon)$$

$$\text{when } t=0 \quad 6 = -2a \sin \epsilon$$

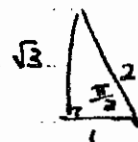
$$a \sin \epsilon = -3 \quad (2)$$

1 mark

$$(2) \div (1) \quad \frac{a \sin \epsilon}{a \cos \epsilon} = \frac{-3}{\sqrt{3}} = -\sqrt{3}$$

$$\therefore \tan \epsilon = -\sqrt{3}$$

$$\therefore \epsilon = -\frac{\pi}{3}$$

(1 mark)
(ECF awarded)now get a

$$\sqrt{3} = a \cos \epsilon \quad (1)$$

$$\sqrt{3} = a \cdot \cos\left(-\frac{\pi}{3}\right)$$

$$\sqrt{3} = a \times \frac{1}{2}$$

$$\therefore a = 2\sqrt{3}$$

$$\therefore x = 2\sqrt{3} \cos\left(2t - \frac{\pi}{3}\right)$$

(1 mark)

(ECF awarded)

12b

OR

$$\frac{1}{2}v^2 = \int -4x dx$$

$$\frac{1}{2}v^2 = -2x^2 + C$$

$$v^2 = -4x^2 + C \quad \text{when } x = \sqrt{3}, v = 6$$

$$6^2 = -4(\sqrt{3})^2 + C$$

$$36 = -12 + C$$

$$36 = -12 + C$$

$$\therefore C = 48$$

$$\therefore v^2 = -4x^2 + 48$$

$$v^2 = n^2(a^2 - x^2)$$

$$v^2 = 4(12 - x^2)$$

$$\therefore a^2 = 12$$

$$a = \sqrt{12} = 2\sqrt{3}$$

$$x = 2\sqrt{3} \cos(2t + \alpha)$$

$$\text{when } t = 0, x = \sqrt{3}$$

$$\sqrt{3} = 2\sqrt{3} \cos \alpha$$

$$\frac{1}{2} = \cos \alpha \quad \therefore \alpha = \frac{\pi}{3}$$

$$\rightarrow \text{note } \ddot{x} = -4\sqrt{3} \sin(2t \pm \frac{\pi}{3})$$

check $x = 6, t = 0 \quad 6 = -4\sqrt{3}(\sin -\frac{\pi}{3})$

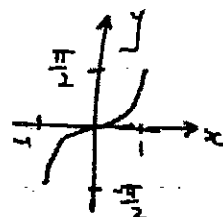
$$\therefore x = 2\sqrt{3} \cos(2t - \frac{\pi}{3})$$

due to $\frac{s/A}{T/C}$

(+ only
error &
process
good) 2/2

12c $y = \frac{1}{2} \sin^{-1}(1-3x)$

$y = \sin^{-1}x$ domain $-1 \leq x \leq 1$
range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

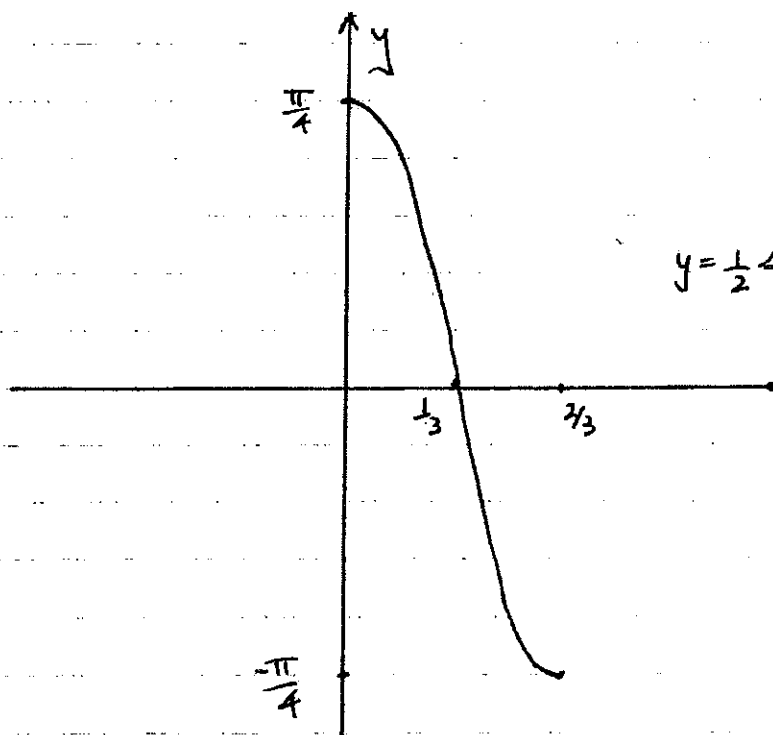


domain $-1 \leq 1-3x \leq 1$
 $-2 \leq -3x \leq 0$
 $\frac{2}{3} \geq x \geq 0$

$\therefore 0 \leq x \leq \frac{2}{3}$

1 mark

range is half normal range $\therefore \frac{\pi}{4} \leq y \leq \frac{\pi}{4}$



$y = \frac{1}{2} \sin^{-1}(1-3x)$

(1 mark)
graph.

12 di $y = \frac{1}{2}(p+q)x - apq$

sub $(0, 5a)$

$$5a = \frac{1}{2}(p+q) \times 0 - apq$$

$$(5a = -apq) \div -a$$

$$pq = -5$$

12d

ii

$$x + py = 2ap + ap^3 \rightarrow x = 2ap + ap^3 - py$$

$$x + qy = 2aq + aq^3$$

↓

$$-py + 2ap + ap^3 + qy = 2aq + aq^3$$

$$y(q-p) = 2aq - 2ap + aq^3 - ap^3$$

$$y(q-p) = 2a(q-p) + a(q^3 - p^3)$$

$$y(q-p) = 2a(q-p) + a(q-p)(q^2 + pq + p^2)$$

$$y = 2a + a(q^2 + pq + p^2)$$

$$y = a(p^2 + pq + q^2 + 2) \quad (1 \text{ mark})$$

sub. into $x = 2ap + ap^3 - py$

$$x = 2ap + ap^3 - pa(p^2 + pq + q^2 + 2)$$

$$x = 2ap + ap^3 - ap^3 - ap^2q - apq^2 - 2ap$$

$$\therefore x = -apq(p+q) \quad (1 \text{ mark})$$

(= 1 mark if valid attempt but mess up y and x)

12d iii

use $pq = -5$

$$x = -apq(p+q)$$

$$\therefore p+q = \frac{-x}{apq} = \frac{-x}{-5a} = \frac{x}{5a}$$

$$y = a(p^2 + pq + q^2 + 2)$$

$$y = a(p^2 + q^2 - 5 + 2)$$

$$y = a[(p+q)^2 - 2pq - 3] \quad (1 \text{ mark})$$

$$y = a\left[\left(\frac{x}{5a}\right)^2 - 2x - 5 - 3\right]$$

$$y = a\left(\frac{x^2}{25a^2} + 7\right)$$

$$y = \frac{x^2}{25a} + 7a \quad (1 \text{ mark})$$

put into form $(x-h)^2 = 4a(y-k)$

$$y - 7a = \frac{x^2}{25a}$$

$$25a(y - 7a) = x^2$$

(1 mark if
worked
forward)

Markers Feedback X1 Trial Q12

Polynomials,

- a was very well done by most students
a few made minor errors, but most
got full marks.

S4M

- b many students had trouble with part ii

→ when you need to get the angle 'd'

it is easier to get 2 equations ① from x
and ② from x to get tan d, then get a.

→ many did $\frac{1}{2}v^2 = \int -4x dx$

this could enable them to get 'a' easily

but was more challenging to get d, as have to use $v=0, t=0$.

→ some integrated inverse trig functions doing way too much work for the marks.

Inverse Trig

- c most calculated the domain & range correctly
However, about 30% of students showed room
for improvement showing insufficient skills.

→ Many drew the graph incorrectly - this is a shame
as substituting $x=0$ would quickly confirm
where the graph lay.

Parametrics

- d i done very well

ii most did this question well, about 20% of
students need to revise their Parametrics work
on finding intersection of 2 points. students should
show $q^3 - p^3 = (q-p)(q^2 + qp + p^2)$ when breaking down
difference of 2 cubes, to ensure they get all marks.

iii about 40% of students struggled.

they need to remember the purpose is to eliminate

p and q → using $pq = -5$, $p+q = \frac{x}{5a}$
and $p^2 + q^2 = (p+q)^2 - 2pq$

Q13

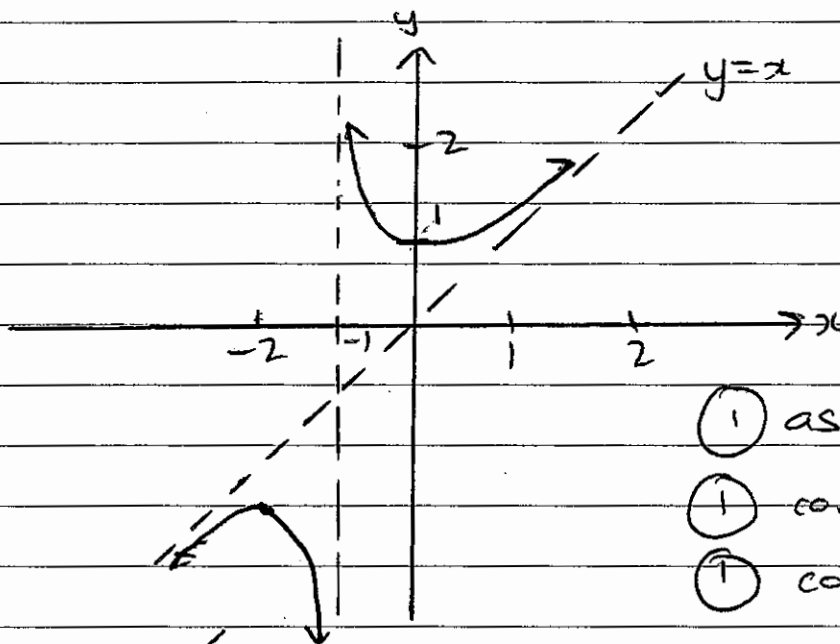
$$a) i) \frac{x^2 + x + 1}{x+1} = \frac{x(x+1) + 1}{x+1} \quad x \neq -1$$

$$= x + \frac{1}{x+1}$$

① mark correct algebraic method
working from LHS $\rightarrow$ RHS

Many students still work
RHS $\rightarrow$ LHS
not what was asked

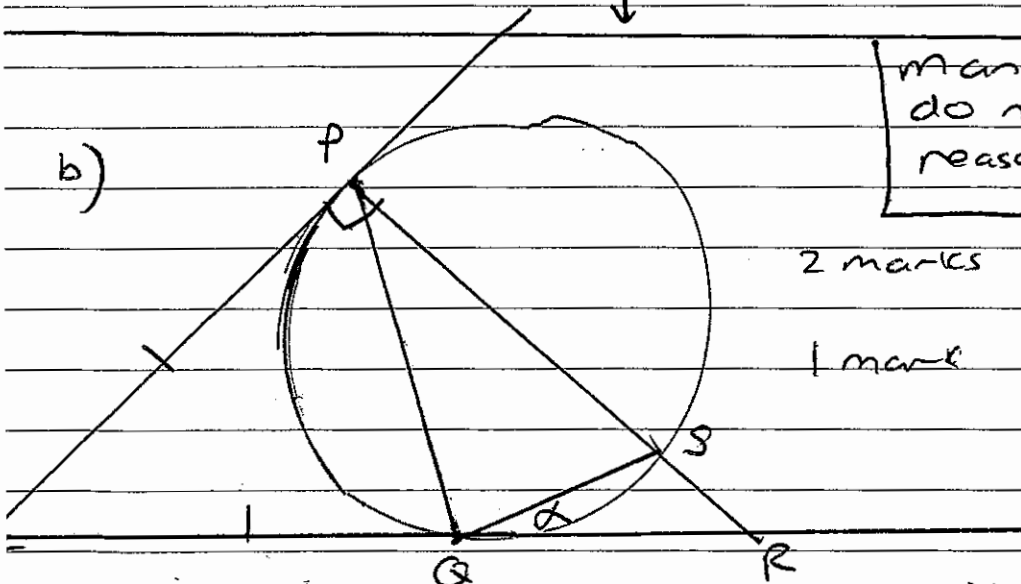
a) ii)



Many students not accurate enough with graph near asymptotes

- ① asymptotes
- ① correct y intercept
- ① correct shape

b)



Many students still do not use correct reasons $\rightarrow$ see syllabus!

2 marks 2 correct reasons that lead to proof
1 mark one correct reason

$\angle TPS = \pi/2$ (tangent to a circle is perpendicular to the radius drawn to point of contact)

✓ ①

$\angle SQR = \angle SPQ = \alpha$ (angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment) ✓ ①

$\therefore \angle TPQ = \pi/2 - \alpha$

b) iii) In ΔTPQ

$TP = TQ$ (tangents to a circle from an external point are equal)

$\therefore \angle TPQ = \angle TQP$ (equal angles opposite equal sides in isosceles ΔTPQ)

$\therefore \angle PTQ = \pi - \left(\frac{\pi}{2} - \alpha + \frac{\pi}{2} - \alpha\right)$ (angle sum of $\Delta TPQ = \pi$)

$\therefore \angle PTQ = 2\alpha$

* 2 marks all correct reasons

✓ 1 mark one incorrect reason

Generally well done

but many still using incorrect reasoning or very inefficient methods

c) ii) $\ddot{x} = 18x(x^2 + 1)$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 18x^3 + 18x$$

$$\therefore \int \frac{d}{dx} \left(\frac{1}{2} v^2 \right) dx = \int 18x^3 + 18x dx$$

✓ ① must have $dx!$

$$\therefore \frac{1}{2} v^2 = \frac{18x^4}{4} + \frac{18x^2}{2} + \frac{C}{2}$$

$$\therefore v^2 = 9x^4 + 18x^2 + C$$

when $x=0, v=3 \rightarrow 9 = C$

$$\therefore v^2 = 9x^4 + 18x^2 + 9$$

$$v^2 = 9(x^4 + 2x^2 + 1) \\ = 9(x^2 + 1)^2$$

but when $x=0, v=3 \rightarrow v > 0$

$$\therefore v = 3(x^2 + 1) \quad \checkmark \text{ ① must have reason for excluding negative}$$

* Many students did not see the factorisation

* Many gave no reason for ignoring the negative and lost a mark

↑
many don't indicate what they are integrating w.r.t.

(13)

c) ii) $v = \frac{dx}{dt} = 3(x^2 + 1)$

$$\therefore \frac{dt}{dx} = \frac{1}{3(x^2 + 1)}$$

$$\therefore \int \frac{dt}{dx} \cdot dx = \frac{1}{3} \int \frac{1}{x^2 + 1} \cdot dx$$

✓ (1) setting up to apply standard integral

$$\therefore t = \frac{1}{3} \tan^{-1}(x) + c$$

when $t = 0, x = 0 \rightarrow c = 0$

$$\therefore t = \frac{1}{3} \tan^{-1}(x)$$

$$\therefore x = \tan(3t)$$

✓ (1) successfully rearranging to achieve solution

Many failed to recognise standard integral

Some left answer as $t = \frac{1}{3} \tan^{-1}(x)$
 $\rightarrow$ look at question

d) $\sqrt{3} \sin 3x - \cos 3x = R \sin(3x - \alpha)$

$$\therefore R = \sqrt{(\sqrt{3})^2 + 1} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \rightarrow \alpha = \pi/6$$

✓ (1) for correct α

✓ (1) for correct R

$$\therefore \sqrt{3} \sin 3x - \cos 3x = 2 \sin(3x - \pi/6)$$

$$\therefore \sqrt{3} \sin 3x - \cos 3x = 2 \sin(3x - \pi/6) = 0$$

$$\therefore 2 \sin(3x - \pi/6) = 0$$

$$\therefore \sin(3x - \pi/6) = 0$$

$$\therefore 3x - \pi/6 = n\pi$$

$$3x = n\pi + \pi/6$$

$$x = \frac{n\pi}{3} + \frac{\pi}{18}$$

✓ (1)

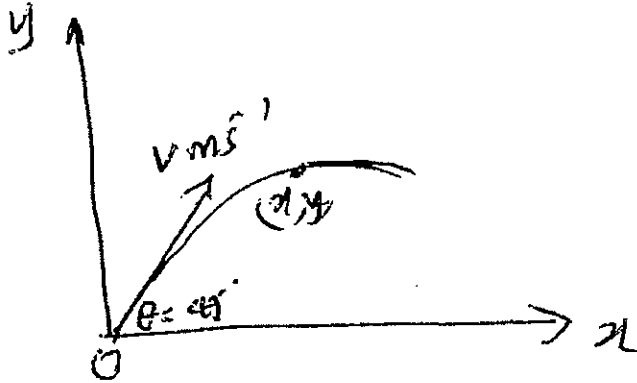
Students who used general

solⁿ formula $\sin \theta = a \rightarrow \theta = n\pi + (-1)^n \sin^{-1} a$

failed to recognise that $a = 0$

Q14

a) (i)



$$\rightarrow \ddot{x} = 0$$

$$\int \frac{dx}{dt} \cdot dt = \int 0 dt$$

$$\dot{x} = C$$

$$t=0, \dot{x} = v \cos \theta$$

$$\dot{x} = v \cos \theta$$

$$\int \frac{dx}{dt} \cdot dt = \int v \cos \theta dt$$

$$x = v \cos \theta \cdot t + D$$

$$t=0, x=0 \therefore D=0$$

$$\therefore x = v \cos \theta \cdot t$$

$$\text{as } \theta = 45^\circ$$

$$\boxed{x = \frac{v t}{\sqrt{2}}}$$

(1 mark)



$$\ddot{y} = -g$$

(1 mark)

$$\int \frac{dy}{dt} \cdot dt = \int -g dt$$

$$\dot{y} = -gt + E$$

$$t=0, \dot{y} = v \sin \theta$$

$$\therefore E = v \sin \theta$$

$$\dot{y} = -gt + v \sin \theta$$

$$\int \frac{dy}{dt} \cdot dt = \int -gt + v \sin \theta dt$$

$$y = -\frac{1}{2} g t^2 + v \sin \theta \cdot t + F, \theta = 45^\circ$$

$$t=0, y=0 \Rightarrow F=0 \therefore \boxed{y = -\frac{1}{2} g t^2 + \frac{v t}{\sqrt{2}}}$$

Many students did not clearly show:

(i) Integrating with respect to what?

(ii) evaluating the constants

(iii) the direction of $\ddot{y}$ and $\dot{y}$ as they are vector quantities.

Q14 a(iii)

$$x = \frac{vt}{\sqrt{2}} \Rightarrow t = \frac{\sqrt{2}x}{v} \rightarrow 1 \text{ mark}$$

$$y = \frac{vt}{\sqrt{2}} - \frac{1}{2}gt^2$$

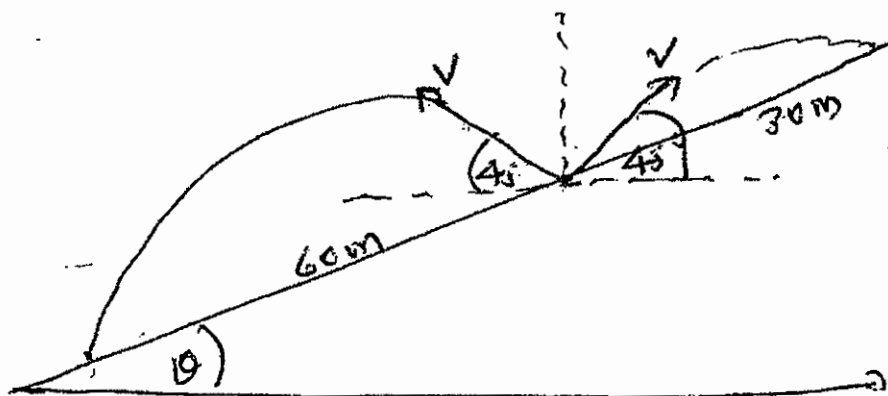
$$= \frac{v}{\sqrt{2}} \cdot \frac{\sqrt{2}x}{v} - \frac{1}{2}g \cdot \frac{2x^2}{v^2}$$

$$= x - \frac{gx^2}{v^2}$$

(1 mark)

most got this correctly

(iii)



Applying the result $y = x - \frac{gx^2}{v^2}$ from a(ii),

For the downhill motion

$$-60 \sin \theta = 60 \cos \theta - \frac{g}{v^2} (60 \cos \theta)^2$$

$$(i) \quad 60 \cos \theta + 60 \sin \theta = \frac{g}{v^2} (60 \cos \theta)^2 \quad (1)$$

For the uphill motion

$$30 \sin \theta = 30 \cos \theta - \frac{g}{v^2} (30 \cos \theta)^2$$

$$(ii) \quad 30 \cos \theta - 30 \sin \theta = \frac{g}{v^2} (30 \cos \theta)^2 \quad (2)$$

$$(1) \div (2) \quad \frac{2(\cos \theta + \sin \theta)}{(\cos \theta - \sin \theta)} = 4 \quad (1 \text{ mark})$$

$$\Rightarrow 3 \sin \theta = \cos \theta$$

$$\Rightarrow \tan \theta = \frac{1}{3} \Rightarrow \theta = \tan^{-1} \left(\frac{1}{3} \right)$$

many students had ~~different~~ ~~first~~ ~~this~~ experienced difficulty with this point

Could not get the equations (1) and (2) with correct signs, again because of not paying attention to the direction

Q14(b)

S = surface Area of the bubble
 V = volume of bubble

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$S = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{ds} \cdot \frac{ds}{dt}$$

$$= 4\pi r^2 \cdot \frac{1}{8\pi r} \cdot (1.9)$$

$$= \frac{1.9r}{2}$$

When $r = 0.6 \text{ mm}$,

$$\frac{dV}{dt} = \frac{1.9 \times 0.6}{2}$$

$$= \underline{\underline{0.57 \text{ mm}^3/\text{s}}}$$

Many got the
current but
some did not
show
clear
working.

Q14 (c) $\cos x - \cos(x+2\theta)$

$$= \cos[(x+\theta)-\theta] - \cos[(x+\theta)+\theta] \quad (1 \text{ mark})$$

$$= \cos(x+\theta)\cancel{\cos\theta} + \sin(x+\theta)\cancel{\sin\theta} - [\cos(x+\theta)\cancel{\cos\theta} + \sin(x+\theta)\cancel{\sin\theta}]$$

$$= \underline{2 \sin(x+\theta) \sin\theta} \quad (1 \text{ mark})$$

Many got this wrong.
Any relevant method v.l.

$$\sum_{r=1}^n \sin(2r-1)\theta = \frac{1 - \cos 2n\theta}{2 \sin \theta}$$

For $n=1$ LHS = $\sin(2-1)\theta = \sin\theta$

$$RHS = \frac{1 - \cos 2\theta}{2 \sin \theta} = \frac{2 \sin^2 \theta}{2 \sin \theta} = \sin \theta$$

$$\Rightarrow \text{LHS} = \text{RHS}$$

(i.e.) The proposition is true for $n=1$ $\rightarrow (1 \text{ mark})$

Assume true for $n=k$

$$(1P) \quad \sum_{r=1}^k \sin(2r-1)\theta = \frac{1 - \cos 2k\theta}{2 \sin \theta}$$

For $n=k+1$ to prove that

$$\sum_{r=1}^{k+1} \sin(2r-1)\theta = \frac{1 - \cos 2(k+1)\theta}{2 \sin \theta} \quad \left. \begin{array}{l} \\ \end{array} \right\} 2 \text{ marks}$$

$$\text{LHS} \quad \sum_{r=1}^{k+1} \sin(2r-1)\theta = \sum_{r=1}^k \sin(2r-1)\theta + \sin[2(k+1)-1]\theta$$

Using the assumption

$$= \frac{1 - \cos 2k\theta}{2 \sin \theta} + \sin[2k+1]\theta$$

$$+ \sin[2(k+1)-1]\theta$$

$$= \frac{1 - \cos 2k\theta + 2 \sin \theta \sin(2k+1)\theta}{2 \sin \theta}$$

$$\text{Let } x = 2k \quad \Rightarrow \frac{1 - \cos x + 2 \sin \theta \sin(x+1)\theta}{2 \sin \theta}$$

Let $x = 2k\theta$

Some students
struggled as
they could not
use the result
from previous
part.

$$= \frac{1 - \cos x + 2 \sin \theta (x + \theta)}{2 \sin \theta}$$

$$= \frac{1 - \cancel{\cos x} + \cancel{\cos x} - \cos(x + 2\theta)}{2 \sin \theta}$$

$$= \frac{1 - \cos(x + 2\theta)}{2 \sin \theta}$$

$$= \frac{1 - \cos[2(k+1)\theta]}{2 \sin \theta}$$

$$= x + \theta$$

(1 mark)

∴ The proposition is true for $n = k+1$.
The proposition is true for $n=1$ and if it is true
for $n=k$ then it is true for $n=k+1$.
Therefore by the method of mathematical induction,
it is true for all positive integers